

Extreme risk spillovers in commodity markets: New insights from multilayer networks in the frequency domain

1. Introduction

In recent years, commodity markets have been exposed to a record number of shocks, highlighting the need to capture time-varying, non-linear risk transmissions. The COVID-19 pandemic, the Russia-Ukraine conflict, and other geopolitical conflicts have initiated widespread falls and spikes in volatilities in energy, metal and agricultural prices. Research reports a “high increase in lower and upper tail contagiousness” amongst commodities since COVID-19 (Qiao and Han [2023](#)). In particular, tail-based contagion networks expose significant sectoral clustering: e.g., energy and metal commodities became highly interconnected, whereas agricultural commodities (especially softs) were relatively less contagious and even offered diversification benefits to investors. These findings demonstrate that extreme events alter the co-movement pattern, with commodity shocks spreading unevenly and unexpectedly throughout the market. Risk managers and policymakers need to grasp this intricate, time-varying structure of linkages – particularly in times of crisis.

Standard models of connectedness (such as simple correlations or the Diebold-Yilmaz generalized variance decomposition (GFEVD) framework) are limited to average linkages and do not consider co-movements that can differ in times of crisis. For example, peripheral measures that capture average linked decisions tend to fail in crisis times, especially within tail events in return distributions (Farid, S et al. [2022](#)). Empirical studies show that average-linkage models may downplay the interconnectedness during stress events and fail to account for the extreme dependence exhibited during market extremes. Similarly, standard portfolio theory relies on stable correlations and normally distributed returns (Naifar, N., & Alhashim, M., [2025](#)). During stress events, it is common for portfolios to obey tail dependence and relate in a way that leads to “behavioural spillovers, tail dependence, and cross-asset contagion, which erodes diversification” (Shahzad et al. [2025](#)). These observations necessitate tail-based and frequency-based network approaches that can capture non-linear, quantile-specific spillovers and the persistence of shocks across various horizons.

One of the ways to do this is through the partial correlation-based connectedness framework presented by [Shahzad et al. \(2025\)](#). The framework shows a network of (conditional) dependencies among commodity returns, which are estimated separately at different quantiles. This framework captures the connection between all the commodities considered, and the limitation of traditional correlation measures is that they do not capture directional interactions. It is very powerful in how it models conditional dependencies by controlling for the influence of all other assets, and allows the user to examine direct, directional connections among commodities. Shahzad et al. show that their method gives extreme quantile-based partially-connected networks meaningfully larger connectedness ds when connectedness is extreme. During a connectedness spike at higher quantiles, only the partial-correlation index singles out the energy commodities; in contrast, the limited bursts of significant interconnections

among a few commodity returns over time are lost in other measures of connectedness. This suggests that the partial correlation-based connectedness quantile networks are robust against model misspecifications while capturing very different conditional and static (mean-based) measures of connectedness. This approach to tail-sensitive connectedness serendipitously reveals when the asset pricing model is mistaken for some assets, or under certain conditions, where connected commodities do not conform to the more symmetric relationship implied by linear models. For example, the authors state that the ‘implied network’ is unstable during turbulent market conditions. This feature is not observed in the average-linkage connectivity models applied to the same commodities. Also, the partial correlation measure allows for constructing market-ready portfolios by forming a portfolio that minimizes the partial correlation (connectedness). In summary, the evidence from [Shahzad et.al \(2025\)](#) supports the idea that partial correlation-based networks are particularly suited for tracking extreme spillovers of connected commodity returns because the model includes a defined notion of tail-dependent, directional, co-movements of conditional asset dependence.

Frequency-domain multi-layer networks also provide another, complementary perspective on risk transmissions among commodities. Linkages among commodities can be thought of in terms of horizons: some shocks move through markets rapidly, while others are felt over a much longer time horizon. A recent “time-varying multi-layer network” approach separately constructs three network layers (short, medium, and long-term frequencies). In this approach, each commodity appears as a node in each layer, and within-layer links capture spillover effects between sub-commodities at a specific frequency. [Zhou et al. \(2024\)](#) implement this design in an analysis of commodity futures and find: “Robust short-term volatility spillovers across commodity markets from stock index and crude oil markets reflect rapid communication of high-frequency shocks”

However, they find that crisis periods shift the frequency spectrum: during a crisis, for instance, long-run connectedness becomes more prevalent, meaning the shocks were able to persist during a crisis phase effectively. Again, a “multi-layer frequency network” allows for analysis of permanent shocks versus transitory shocks. The edge structure also varies across frequency layers; for example, their results found cocoa futures emerged to be a “hedge” in specific frequency layers, while inflation risk produces more long-run spillover and risk. Ultimately, frequency-domain networks begin to clarify which commodity shocks die out quickly and which shocks exhibit persistent features that contribute to systemic linkages. Thus, frequency-domain networks present a more nuanced structural viewpoint on systemic linkages when compared to aggregate models.

Combining the two modalities affords a more complete understanding of commodity transmission of risk. The partial-correlation networks capture extreme directional co-movements, while the frequency multi-layer networks allow for understanding how those extreme co-movements evolve over different horizons. For example, it can help illuminate whether a tail shock in oil will immediately transmit to agricultural prices or only contribute to long-term metal cycles. This is tangible value. With tail-networks, investors can take advantage of differing developments between sectors, e.g., by rotating hedges using soft commodities during energy crises. Likewise, a regulatory authority or a risk manager can develop maps of the network in order to identify important risk transmitters. Some studies in commodity transmission highlight hub commodities, e.g. copper, that disproportionately carry tail shocks and merit special attention. The partial-corr/RW towards diversified portfolios show already

that using network-associated portfolios resulted in improved risk compared to alternative portfolios. The frequency-layered multi-layer network can assist regulators in constructing regulatory stress tests, particularly around how persistent shocks tend to be. In short, a joint network analysis based on tail/shock and frequency is a multi-layer commodity network that could also be used to think about commodity spillovers, which is also time-varying and directional, and to help inform diverse portfolio construction, oversight of systemic risk, and even policy..Given recent market turbulence, such an integrated analysis is timely. This paper therefore replicates and extends (Shahzad et al.[2025](#)) to commodity portfolios, applying their partial correlation connectedness in the tails. It then augments this with a multi-layer frequency-domain network spanning energy, metals and agriculture. By doing so, we aim to illuminate both the conditional tail-dependencies and the horizon-dependent spillover dynamics that shape extreme risk transmission in commodity markets.

2. Literature Review

There has been an increase in the study of how risks spread between different commodity markets, especially during tough times like financial crises or when governments make uncertain policy decisions.

Many studies now use advanced statistical methods and network analysis to understand how bad events in one market can affect others. This research is important because it helps us understand:

- How price shocks move from one commodity to another
- Why are some markets more connected than others during crises
- How these connections affect investment decisions
- Whether the entire financial system becomes unstable when problems spread

This knowledge helps investors build better portfolios, helps regulators spot potential system-wide risks, and gives policymakers better tools to prevent or manage financial crises.

Connectedness and Tail Risk in Commodity Markets

The connectedness of commodity returns varies across different market conditions. During crisis periods, certain methods better detect heightened market stress and identify structural vulnerabilities. One approach that effectively captured commodity shocks during crises is the total connectedness measure based on partial correlation that decomposes volatility regimes. It is noted that constructing portfolios based on this approach improves risk-adjusted returns, especially when compared to traditional methods that do not account for changes between low and high volatility periods (Shahzad et al., [2025](#)).

Regional market structures influence relationships between stocks and commodities, indicating that diversification can help at times. Commodities, such as precious and industrial metals, are found to act

as safe havens or hedges during periods of financial uncertainty, but these roles are not consistent across all markets. (Sajid et al., [2020](#)).

In new asset markets, such as cryptocurrencies, the transmission of extreme risks follows particular patterns. Some are transmitters while others are receivers of shock spillovers. Classifying cryptocurrencies into speculative and technical groups makes it possible to understand how tail risks spread (Ahelegbey et al., [2021](#)).

Risk transmission is not limited to traditional assets but applies to new investment avenues such as impact investing. Calm or crisis periods, irrespective, impact investments transmit shocks across markets, while agricultural commodities absorb these shocks. Portfolios that combine both types of assets benefit from enhanced hedging effectiveness (Banerjee et al., [2024](#)).

Uncertainty, Sentiment, and Economic Shocks

Behavioural factors, such as investor sentiment and market uncertainty, push the reactions of agricultural commodity returns during the initial pandemic wave, highlighting the importance of psychological influences in crisis periods (Akyildirim et al., [2022](#)). Similarly, contagion between major economies spiked during the pandemic, with conditional correlations between stock returns of financial and non-financial firms rising, particularly among financial firms. This suggests that crises force interdependencies and call for robust regulatory oversight (Boubaker et al., [2021](#)).

Economic policy uncertainty, for example, U.S. monetary policy, has been identified as a major driver of spillovers between oil and commodity currency markets. These effects are more in the short-term among G10 currencies and were further intensified after the Global Financial Crisis of 2008 (Albulescu et al., [2019](#)). Uncertainties increase correlations between stock and commodity returns, especially in weak economic regions. (Badshah et al., [2019](#)).

Systemic Risk, Policy Spillovers, and Crisis Dynamics

Policy shocks and global integration play a central role in driving systemic risk and spillovers across markets. Large-scale interventions like U.S. quantitative easing (QE) supported recovery efforts in both advanced and emerging markets but also introduced risks, with emerging economies experiencing more intense spillovers than advanced ones (Chen et al., [2016](#)). The effects of QE were comparable to conventional monetary policy tools such as rate cuts, inducing capital inflows, raising equity prices, and reducing bond spreads in emerging markets (Peter Tillman, [2016](#)).

Within Europe's sovereign debt markets, feedback loops between banks and governments intensified during the debt crisis, although timely policy interventions helped mitigate systemic risks (Alter et al., [2014](#)).

At a finer level, contagion effects vary across sectors. During the Global Financial Crisis and European Sovereign Debt Crisis, financial services and telecommunications sectors were found to be particularly prone to contagion, especially in core EU countries (Alexakis et al., [2018](#)). Non-financial firms have been shown to be stronger channels of contagion, with bilateral trade relationships and macroeconomic conditions further causing dependencies during global crises. (Akhtaruzzaman et al., [2016](#)).

Commodity Markets, Policy Effects, and Structural Shifts

Commodity markets have also undergone structural shifts caused by persisting trends such as financialization. September 2008 marked the transition of commodities from assets used for crisis hedging to strategic investment assets, a shift due to growing financialization rather than isolated crisis events (Zeno et al., [2015](#)).

Investor expectations regarding economic growth have also evolved following the COVID-19 outbreak, with models showing declines in expected dividend and GDP growth. Policy interventions temporarily boosted long-term expectations, though short-term sentiment remained slightly pessimistic (Neils et al., [2022](#)).

Cross-sectional analyses of stock returns further reveal that global supply chains and trade linkages played an important role in determining firm performance during the pandemic. Firms exposed to China trade or those with weak fundamentals underperformed significantly, while corporate debt and liquidity metrics gained popularity as key value drivers amid market interventions (Stefano R et al., [2020](#)).

Hence, this shows that risks in commodity markets spread in complex ways through multiple channels. Studies reveal that when one market has problems, these issues can quickly leak over to other markets. To protect against this, studies recommend using sound risk management practices backed by diversified portfolios and necessary government regulation during crises to prevent the damage from spreading throughout the financial system.

3. Research Gap

There are still significant gaps in the scope and methodology of the expanding research on risk spillovers in the energy and commodity markets. Much of the current work uses decomposition-based connectedness approaches in the time domain, which are helpful but sometimes fail to capture frequency-dependent dynamics and extreme tail interactions. Moreover, new methodologies like partial correlation-based frameworks that better distinguish between contemporaneous dependence and lagged spillovers (Shahzad et al., [2025](#)) and the application of multi-layer networks in the frequency domain to analyze global energy markets (Jin et al., [2024](#)) have further extended this line of research. However, most of this literature focuses on regional or global commodity markets, frequently combining various economies based on income, demographics or development. The structural and systemic role of just the United States of America is rarely considered. This country remains the largest commodity consumer and one of the most influential producers across multiple sectors. This creates an important gap, as U.S. commodity markets have distinct features — deep financialization, a central role in global price discovery, and heightened sensitivity to macro-financial shocks — that may yield risk spillover dynamics materially different from those documented in international or emerging-market settings. Methodologically, prior research also leaves questions regarding integrating multi-layer frequency-domain analysis with extreme-risk measures. While time-domain studies have shown strong contagion effects during major crises such as the Global Financial Crisis, the COVID-19 pandemic, or the

Russia–Ukraine conflict, they often obscure heterogeneity across short, medium, and long-term horizons. Moreover, Single-layer network models neglect the interaction between frequency bands and provide only a partial picture of systemic transmission. This study addresses these gaps by applying a multi-layer network framework in the frequency domain, conditional autoregressive value-at-risk (CAViAR) measures, and LASSO-VAR-BK spillover estimation to the U.S. commodity market, which remains the largest commodity consumer and one of the most influential producers across multiple sectors. This creates an important gap, as U.S. commodity markets have distinct features — deep financialization, a central role in global price discovery, and heightened sensitivity to macro-financial shocks — that may yield risk spillover dynamics materially different from those documented in international or emerging-market settings.

4. Objectives of Study

The study investigates the return dynamics of 13 selected commodities to understand their risk–return characteristics. It examines interdependencies and spillover effects among these commodities across different layers of analysis. Moreover, it attempts to measure the heterogeneity in out-strength and in-strength across the commodities. Last but not the least, it provides insights into the stability and transmission of shocks within the commodity market network.

5. Data

We use daily closing (settlement) prices expressed in USD of 13 commodities futures, namely Crude Oil, Heating Oil, Gasoline, Corn, Wheat, Soyabeans, Rice, Sugar, Gold, Silver, Platinum, Palladium, Copper. These commodities are grouped into four different sectors, namely energy, agricultural, precious metals, and industrial metals. Price data are downloaded from Bloomberg for the period January 1, 2006, to June 30, 2025, yielding a total of 4909 daily observations for each commodity. Interestingly, the sample period is long enough to cover key and various events and crisis periods, including the global financial crisis 2007–09, the oil price collapse from mid-2014 to January 2016, the COVID-19 pandemic since early 2020, and the ongoing war in Ukraine from 24 February 2022. For each commodity, we compute the daily log return as the natural logarithmic ratio of two consecutive daily prices via the formula.

[Table 1](#) presents the details as well as the descriptive and stationarity statistics of the return series of the sampled commodities for the entire period under study. The mean returns are positive for all commodities. Unsurprisingly, the standard deviation is much larger in energy commodities, reflecting the large swings in energy prices, especially around the peak of the pandemic period in April 2020. Most of the skewness values are negative, except for wheat. The kurtosis values are exceptionally large for the returns of rice and crude oil. All return series deviate from the normal distribution, as shown in the Jarque-Bera statistics. Furthermore, all series are stationary at levels, according to the statistics of the Augmented Dickey-Fuller unit root test.

S. No.	Group	Commodity	Mean	Variance	Skewness	Ex-Kurtosis	JB	ADF
1	Energy	Crude Oil	1.32E-05	7.71E-04	-1.9886	59.6754	7.32E+05***	-41.13***
2		Heating Oil	6.23E-05	5.04E-04	-0.9552	12.1714	3.10E+04***	-39.49***
3		Gasoline	3.63E-05	7.13E-04	-1.2532	22.7669	1.07E+05***	-40.34***

4	Agricultural	Corn	1.36E-04	3.53E-04	-1.0666	14.8286	4.59E+04***	-38.31***
5		Wheat	9.04E-05	4.47E-04	0.2565	3.1688	2.11E+03***	-37.62***
6		Soyabeans	1.08E-04	2.27E-04	-0.763	5.1515	5.90E+03***	-37.66***
7		Rice	9.99E-05	2.67E-04	-2.5286	46.1823	4.41E+05***	-38.62***
8		Sugar	5.83E-05	2.92E-04	-0.7211	5.7471	7.18E+03***	-36.74***
9	Metals	Gold	3.77E-04	1.29E-04	-0.3334	5.2035	5.63E+03***	-36.55***
10		Silver	2.84E-04	4.35E-04	-0.7623	6.3328	8.68E+03***	-35.58***
11		Platinum	6.43E-05	2.70E-04	-0.3223	3.642	2.80E+03***	-37.84***
12		Palladium	2.94E-04	5.12E-04	-0.3155	7.7023	1.22E+04***	-35.31***
13		Copper	1.84E-04	3.00E-04	-0.1456	3.7916	2.96E+03***	-35.89***

Table 1. Details and summary statistics of commodities returns.

Note: This table shows the group, name, and acronym of 13 commodities together with the descriptive statistics and unit root test results of their daily returns for the whole sample period. The sample period spans January 1, 2006 to Jun 30, 2025, with a total of 4909 daily observations for each commodity. The commodity returns are calculated as the natural logarithmic ratio of two consecutive daily prices. The JB is the statistic of the Jarque–Bera test of normality. ADF represents the conventional Augmented Dickey-Fuller unit root test. As usual, *** indicates the rejection of the corresponding null hypothesis at the 1 % significance level.

6. Methodology

6.1 CAViar

We utilize the conditional autoregressive value at risk (CAViaR) model proposed by Engle and Manganelli (2004) to calculate the extreme risk in a country's energy market. This approach has the advantage of modeling the quantile directly rather than the entire distribution. A general CAViaR specification is given by:

$$\text{VaR}_{it}(\theta) = \alpha_0 + \sum_{j=1}^L \alpha_j \text{VaR}_{it-j}(\theta) + \sum_{j=1}^L \beta_j r_{it-j} \quad (1)$$

where $\text{VaR}_{it}(\theta)$ is the value at risk (VaR) of energy market i on day t , L represents the function with a finite number of lagged observations, and r_{it-j} is lagged returns. $\text{VaR}_{it}(\theta)$ is the autoregressive term. There are four models for CAViaR: (1) asymmetric slope, (2) indirect GARCH, (3) symmetric absolute

value, and (4) adaptive. We use the asymmetric slope model to calculate the VaR for the energy market:

$$\sigma_{\alpha}(\alpha) = \sigma_{\alpha 0} + \sigma_{\alpha 1} \sigma_{\alpha-1}(\alpha) + \sigma_{\alpha 2} (\sigma_{\alpha-1})^+ + \sigma_{\alpha 3} (\sigma_{\alpha-1})^- \quad (2)$$

6.2. Measuring the spillover matrix in the frequency domain

To measure risk spillovers between markets at different frequencies, Barunik and Krehlik (2018) proposed a spillover index model based on the spectral representation of the generalized variance decomposition of the vector autoregression (VAR) model, known as the BK spillover index model. When applied to high-dimensional samples, the traditional VAR model faces the “dimensionality dilemma” because there are too many parameters to estimate. The BK spillover index based on the LASSO-VAR model (i.e., the LASSO-VAR-BK model) can solve this problem (Demirer et al., 2018). Therefore, we utilize the LASSO-VAR-BK model to measure the risk spillovers between 20 energy markets at different frequency bands and obtain the spillover matrix in the frequency domain.

Based on the VAR of N energy markets (variables), we construct a vector autoregressive (VAR(p)) model of order p :

$$\alpha_t = \sum_{i=1}^p \Phi_i \alpha_{t-i} + \epsilon_t \quad (3)$$

Where, $\alpha_t = (\alpha_{1t}, \alpha_{2t}, \dots, \alpha_{Nt})'$ is an $N \times 1$ vector representing the VaR values of N energy markets. The order of the VAR model, p, is chosen optimally using the SC criterion. The matrices Φ_i ($i=1,2,\dots,p$) are of dimension $N \times N$, while $\epsilon_t \sim N(0, \Sigma_t)$ denotes the error term.

If the VAR(p) process satisfies the stability condition, it can be rewritten as a moving average (MA) process: $\alpha_t = \sum_{i=0}^{\infty} \Psi_i \alpha_{t-i}$, where each Ψ_i is an $N \times N$ coefficient matrix defined recursively as $\Psi_0 = I$, $\Psi_1 = -\Phi_1$, $\Psi_2 = -\Phi_2 - \Phi_1 \Psi_1$, $\Psi_3 = -\Phi_3 - \Phi_1 \Psi_2 - \Phi_2 \Psi_1$, $\dots$, $\Psi_p = -\Phi_p - \Phi_1 \Psi_{p-1} - \dots - \Phi_{p-1} \Psi_1$.

We employ the Least Absolute Shrinkage and Selection Operator (LASSO) to restrict the number of parameters to be estimated. By introducing a penalty term into the VAR, we obtain the LASSO-VAR model, which reformulates the estimation as an optimization problem.

$$\min_{\Phi} \sum_{t=1}^T \left\| \alpha_t - \sum_{i=1}^p \Phi_i \alpha_{t-i} \right\|_2^2 + \lambda \|\Phi\|_1 \quad (4)$$

where $\|A\|$ represents the Frobenius norm of matrix A, λ is a non-negative penalty term coefficient, and $\|\Phi\|_1$ is the L1 parametrization of the matrix Φ .

Based on Eq. (4), we perform generalized forecast error variance decomposition (GFEVD). For the H-step ahead forecast error variance of variable, the share caused by variable can be expressed as:

$$\theta_{j,k}^{\omega} = \frac{\sum_{h=0}^{L-1} \left(\sum_{l=0}^{L-1} (\Psi_h^{\omega})_{j,k} \right)^2}{\sum_{h=0}^{L-1} \left(\sum_{l=0}^{L-1} (\Psi_h^{\omega})_{j,k} \right)}, \quad j, k = 1, 2, 3, \dots \quad (5)$$

Here, $\sigma_{kk} = (\Sigma)_{kk}$ and Ψ_h denotes the coefficient matrix of the moving average model.

Using the frequency response function $\Psi(e^{-i\omega}) = \sum_h e^{-i\omega h} \Psi_h$, the spectral density $S_X(\omega)$ of X_t at frequency ω can be expressed as:

$$S_X(\omega) = \sum_{h=-\infty}^{\infty} \left(\sum_{l=-\infty}^{\infty} \Psi_h^{\omega} \right) e^{-i\omega h} = \left(\sum_{l=-\infty}^{\infty} \Psi_l^{\omega} \right) \left(\sum_{h=-\infty}^{\infty} \Psi_h^{\omega} \right)^* \quad (6)$$

In this expression, $\Psi(e^{-i\omega})$ is obtained from the Fourier transform of Ψ_h , and $i = \sqrt{-1}$. The spectral density $S_X(\omega)$ describes how the variance of X_t is distributed across different frequencies, making it an essential tool for understanding frequency dynamics.

The spectral density and the GFEVD are combined to form the GFEVD at ω frequency:

$$\theta_{j,k}^{\omega} = \frac{\sum_{h=0}^{L-1} \left(\sum_{l=0}^{L-1} (\Psi_h^{\omega})_{j,k} \right)^2}{\sum_{h=0}^{L-1} \left(\sum_{l=0}^{L-1} (\Psi_h^{\omega})_{j,k} \right)} \quad (7)$$

The normalized spillover share $\theta_{j,k}(\omega)$ is defined as:

$$\widetilde{\theta}_{j,k}^{\omega} = \frac{\theta_{j,k}^{\omega}}{\sum_{k=1}^N \theta_{j,k}^{\omega}} \quad (8)$$

where $\theta_{j,k}^{\omega}$ represents the extreme risk spillover strength transmitted from the k th variable to the j th variable at frequency ω .

Given a frequency band $d_i = (a, b)$, the cumulative extreme risk spillover from the k th variable to the j th variable can be written as:

$$\widetilde{\theta}_{j,k}^{\omega}(d_i) = \int_a^b \widetilde{\theta}_{j,k}^{\omega}(\omega) d\omega \quad (9)$$

where $a, b \in (-\pi, \pi)$ $a < b$.

Using Equation (9), the spillover matrix $C(d_i)$ for N variables over the frequency band d_i can be constructed, as shown in Table below.

Table 1. Spillover matrix in the frequency domain.

	X_1	X_2	$\dots$	X_N
X_1	$\tilde{\theta}_{1,1}(d_i)$	$\tilde{\theta}_{1,2}(d_i)$	$\dots$	$\tilde{\theta}_{1,N}(d_i)$
X_2	$\tilde{\theta}_{2,1}(d_i)$	$\tilde{\theta}_{2,2}(d_i)$	$\dots$	$\tilde{\theta}_{2,N}(d_i)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
X_N	$\tilde{\theta}_{N,1}(d_i)$	$\tilde{\theta}_{N,2}(d_i)$	$\dots$	$\tilde{\theta}_{N,N}(d_i)$

Following Ouyang and Zhou (2023), we divide the frequency spectrum into three distinct bands:

- $d1=(\pi/5,\pi)$, representing the **high-frequency band** corresponding to short-term spillovers of 1–5 days,
- $d2=(\pi/20,\pi/5)$ representing the **medium-frequency band** corresponding to medium-term spillovers of 5–20 days, and
- $d3=(0,\pi/20)$, representing the **low-frequency band** corresponding to long-term spillovers of more than 20 days.

Using these bands, we construct the spillover matrix in the frequency domain, denoted as $C(d_i)$, where $i=1,2,3$ corresponds to the short-term, medium-term, and long-term spillover matrices, respectively.

6.3. Construction of multilayer networks in the frequency domain

On the basis of the spillover matrix $C(d_i)$ ($i=1,2,3$), we construct multilayer networks in the frequency domain $\Omega=(G(d1),G(d2),G(d3))$, where $G(d1)$, $G(d2)$ and $G(d3)$ represent the short-term extreme risk spillover layer, the medium-term extreme risk spillover layer, and the long-term extreme risk spillover layer, respectively. $G(d_i)=G(V,C(d_i))$ where $V=\{1,2,\dots,N\}$ is the set of nodes, and N is the number of energy markets. $C(d_i)$ is the set of edges or the spillover matrix of the frequency band d_i . The element $\theta'_{j,k}(d_i)$ in $C(d_i)$ is a directed and weighted edge that represents the extreme risk spillover strength from the k th variable to the j th variable over the frequency band d_i . Therefore $\Theta'_{j,k}(d_i) \neq \theta'_{k,j}(d_i)$.

Multilayer networks in the frequency domain of this paper have the same set of nodes in each layer, but the nodes in different layers are linked at varying frequencies. We construct each layer based on the LASSO-VAR-BK model, which implies that each layer is a fully connected network containing redundant or noisy information. We filter such information using the threshold method to retain the most important information in each layer. We obtain the filtered network by taking the average value of the spillover matrix $C(d_i)$ as a threshold and removing edges smaller than the threshold. Notably, the multilayer networks in the frequency domain Ω we constructed are static networks. To explore dynamic extreme risk spillovers in international energy markets, we use the rolling window approach to construct dynamic multilayer networks in the frequency domain Ω'_t , where $t=1,2,\dots,T$.

6.4. Network measures

To analyze the topological properties of multilayer networks in the frequency domain, we introduce two types of measures: single-layer network measures and multilayer network measures. Single-layer network measures are computed based on any individual layer, while multilayer network measures are computed based on two or three layers.

6.4.1. Single-layer network measures

(1) System-level measures

To quantify the connectedness of a single-layer network, following Wang et al. (2021), we utilize average connectedness strength (ACS) and average node degree (AND). ACS measures the average total spillovers from each energy market to the entire international energy system. A single-layer network with a larger ACS value has a higher total connectivity. The ACS of layer d_i is given by:

$$ACS_{d_i} = \frac{1}{N} \sum_{j=1}^N \sum_{k=1, k \neq j}^N \widetilde{\theta'_{j,k}(d_i)} \quad (10)$$

Where $\theta'_{j,k}(d_i)$ denotes extreme risk spillover strength from country k to j on layer d_i .

AND represents, on average, the number of edges connected to each country. A single-layer network with a larger AND value possesses more edges and more complex risk transmission paths. The AND of layer d_i is given by:

$$AND_{d_i} = \frac{1}{2N} \sum_{j=1}^N k_j^{(d_i)} \quad (11)$$

where $k_j^{(d_i)}$ represents the degree of country (node) j on layer d_i and is defined as the sum of the number of edges both accepted and emitted by country j .

6.4.2. Multilayer network measures

(1) System-level measures

Following Musmeci et al. (2017) and Wang et al. (2021), we use two multilayer measures to quantify the edge structure similarity of multilayer networks in the frequency domain: the uniqueness edge ratio (UER) and the average edge overlap (AEO).

The UER of layer d_i reflects the uniqueness of edges in this layer, which is defined as the proportion of edges on layer d_i that do not exist on other layers:

$$UER_{d_i} = \frac{1}{N} \sum_{j=1, j \neq d_i}^D \prod_{k=1, k \neq j}^D (1 - \frac{E_{j,k}}{N}) \quad (19)$$

where $a_{jk}^{d_i} = \text{sign}(\theta'_{j,k}(d_i))$. If $\theta'_{j,k}(d_i) \neq 0$, then $a_{jk}^{d_i} = 1$. $\square_{\square, \square=1, \square \neq \square} = \sum_{\square, \square=1, \square \neq \square} \square_{\square, \square}^{\square}$ is the number of edges on layer d_i . M denotes the number of layers of multilayer networks in the frequency domain, and $M=3$ in this paper. $UER^{d_i} \in [0,1]$. A larger UER^{d_i} indicates more unique edges on layer d_i .

AEO is the average number of layers on which each edge will appear:

$$\square_{\square, \square} = \frac{1}{\square} \sum_{\square, \square=1, \square \neq \square} \square_{\square, \square}^{\square} \quad (20)$$

where $\square = \sum_{\square, \square=1, \square \neq \square} [1 - \prod_{\square=1}^{\square} (1 - \square_{\square, \square}^{\square})]$ is the number of edges that appear on at least one layer. The value of AEO is between 1 and 3. AEO=3 represents that the edge structure is identical for three layers, while AEO=1 indicates that each edge appears on just one layer on average.

To measure the similarity in the spillover strength (i.e., edge weight) structure between any two layers, we compute the correlation between the edge weights of layers d_i and d_g . We use the Spearman's coefficient (ρ^{d_i, d_g}) and the Kendall's coefficient (τ^{d_i, d_g}), as defined in Eqs. (21), (22), respectively:

$$\rho^{\square_{\square}, \square_{\square}} = 1 - \frac{6 \sum_{\square} \left(\widetilde{\square_{\square, \square}(\square_{\square})} - \widetilde{\square_{\square, \square}(\square_{\square})} \right)^2}{\square (\square^2 - 1)}, \quad \square_{\square}, \square_{\square} = 1, 2, 3 \quad (21)$$

where $\theta'_{j,k}(d_i)$ and $\theta'_{j,k}(d_g)$ denote the edge weights between country j and country k on layers d_i and d_g , respectively. N is the number of countries. $\rho^{d_i, d_g} \in [-1,1]$, where $\rho^{d_i, d_g}=1$ denotes a completely positive correlation between the spillover strength structures of the given two layers, and $\rho^{d_i, d_g}=-1$ denotes a completely negative correlation. A larger value of ρ^{d_i, d_g} indicates a higher similarity in the spillover strength structure of layers d_i and d_g .

$$\tau^{\square_{\square}, \square_{\square}} = (\square - \square) \frac{2}{\square (\square - 1)} \quad (22)$$

where n stands for the number of edges. The variables P and Q represent the number of concordant and discordant pairs, respectively, among the n edges of the layer d_i and layer d_g . Notably, $\tau^{d_i, d_g} \in [-1,1]$. The meaning of the value of τ^{d_i, d_g} is consistent with that of ρ^{d_i, d_g} .

7. Results

7.1. Risk Dynamics and Systemic Stress (CAViaR and LOO)

The analysis begins by looking at how risk patterns emerge and evolve for individual commodities, using what's called the Conditional Autoregressive Value at Risk (CAViaR) method. To figure out which commodities matter most for overall market stability, we apply a Leave-One-Out (LOO) approach. Essentially, this technique asks: "What happens to the system's total risk if we take one commodity out of the picture?" The answer tells us how much that particular commodity adds to the market's vulnerability when prices take a sharp dive.

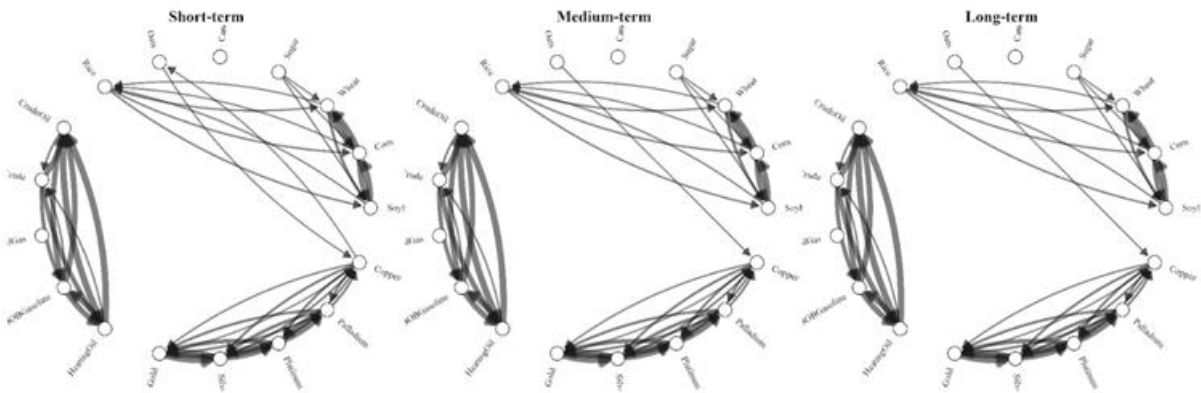


- **Crisis Peaks and Synchronization:** When we look at the time series of average LOO values shown in the figure above, we notice dramatic spikes that happen across multiple commodities at the same time (Zhou, 2024); a clear sign that the entire market is under stress. These surges aren't random. They align almost perfectly with major global disruptions, particularly the Global Financial Crisis that hit in 2008–2009 and the market chaos that erupted when COVID-19 first struck in early 2020. These overlapping peaks suggest that large-scale financial and health crises quickly spread through commodity markets, triggering collective stress and sharply increasing the likelihood of extreme losses across the sector.
- **Structural Risk Drivers (Commodity-Level LOO - Table 7):** The stability and magnitude of risk contributions are confirmed by the top LOO rankings. The results unequivocally show the persistent dominance of **Crude Oil** and refined products (Balli 2019) across all memory layers (short-, medium-, and long-term), solidifying the Energy sector's role as the primary, structural source of individual extreme risk.

Rank	Commodity (Short-term layer)	L O O	Commod ity (Medium	LOO (%)	Commodity (Long-term layer)	LOO (%)
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		(%)	-term layer)			
1	Crude Oil	21.75	Crude Oil	21.88	Crude Oil	21.88
2	Heating Oil	21.06	Heating Oil	21.16	Heating Oil	21.16
3	Silver	20.73	Silver	20.53	Silver	20.52

7.2. Static and Multilayer Network Structure

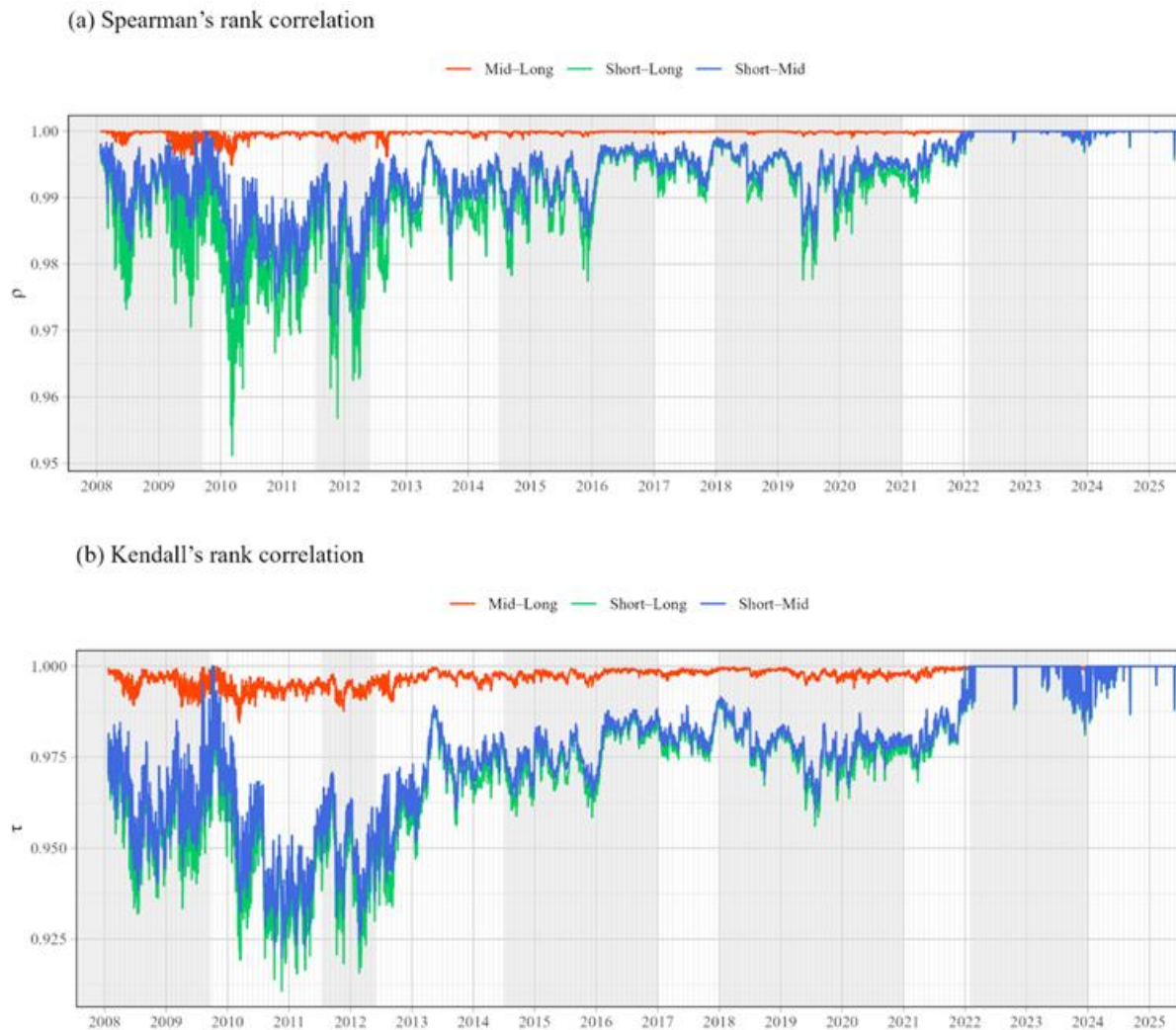


7.2.1 Static Multilayer Network Visualization

When we look at the static multilayer network visualization, we're essentially getting an average snapshot of how risk is organized across the commodity market over different time periods; short-term, medium-term, and long-term. The first thing you'll notice is a pretty clear pattern: the short-term layer is much denser than the others ([Qiang2024](#)) This density means that when we focus on shorter time frames, commodities are far more interconnected with each other.

What's interesting is that no matter which time layer we examine, the same players keep showing up in the spotlight. Energy commodities like Crude Oil and Brent Crude, along with metals like Gold and Copper, consistently sit at the heart of the network. This isn't just coincidence; it shows us that these commodities play a crucial role in connecting the broader market together.

7.2.2 Inter-Layer Structural Similarity



The structural relationship between the network layers is formally assessed using **Spearman and Kendall rank correlation coefficients**, which measure the similarity of the commodities' risk ranking (centrality) across different memory windows. ([Dai 2023](#))

- **High Similarity (rho):** Both correlation coefficients between the short-, medium-, and long-term layers are consistently high (often above 0.8) across the entire sample period.
- **Implication:** This exceptionally high correlation confirms that the fundamental structure of risk relationships is highly rigid and stable across time horizons. This relative systemic importance of the commodities remains largely the same irrespective of whether risk is measured over 22 days or 252 days.

One area of slight divergence is found in sectoral centrality rankings in some papers, where agricultural commodities show weaker network integration and variability in centrality ([saqib 2019](#)), while our analysis maintains the dominance of energy and metals throughout.

7.3. Dynamic Systemic Risk of Commodity Groups

We wanted to see how systemic risk in the commodity network changes over time, so we tracked network centrality measures across three different time horizons: short-term (22 days), medium-term (66 days), and long-term (252 days). Our focus landed on two key indicators that really matter: Average Closeness Centrality (ACS) and Average Neighbor Degree (AND).

7.3.1 Average Closeness Centrality (ACS)

Here's what ACS actually measures: imagine a shock hits one commodity; how long does it take for that ripple effect to reach every other commodity in the market? ACS captures this phenomenon exactly. It counts the steps needed for risk to travel through the network. Higher the ACS value, the fewer steps required, which means risk is spreading faster and more directly. And when risk spreads that efficiently, we're looking at heightened systemic risk.

- **Memory Window Effect:** Looking at our data across the board; something really clear jumps out: short-term ACS consistently sits higher than medium-term, and medium-term beats long-term every time. This pattern isn't random; it's showing us that the freshest market movements and the most recent shocks are what really get commodities talking to each other right away. New information doesn't just influence the market; it actively creates these instant connections through which risk flows. However, these connections don't last forever. With time, and the market digesting what happened, those transmission channels start to fade and weaken.

(a) Dynamic ACS



- **Crisis Peaks:** If you scan through the ACS timeline, you'll notice some dramatic spikes that really catch your eye, and when you look at the timing, they line up remarkably well with major global financial meltdowns:
 - **2008 Global Financial Crisis (GFC):** During this crisis, the short-term ACS shot up in a big way. Here's what was actually happening: the credit crunch that started in financial markets didn't just stay put; it spread outward and fundamentally rewired how the commodity network operated, making everything far more connected than before. What this meant in practical terms was that risk could now move quickly and easily across all three major commodity sectors (Agriculture, Energy, and Metals). Problems that might have stayed isolated in one area suddenly became threats to the entire system.
 - **2020 COVID-19 Market Turmoil:** This crisis was something else entirely. All three memory windows; short, medium, and long-term; spiked at exactly the same time, and the short-term ACS hit the highest point we've seen in the entire dataset. We're talking about a moment of extreme market panic where the system became so tightly integrated that risk wasn't just spreading fast it was moving almost instantly from one commodity to another.

A difference, however, is seen in Raza et al. (2022), whose dynamic network study finds metals, especially precious metals, sometimes act as early shock absorbers—reducing risk transmission in certain crises, contrary to our finding of uniform risk integration across sectors ([syed 2024](#)).

7.3.2 Average Neighbour Degree (AND)

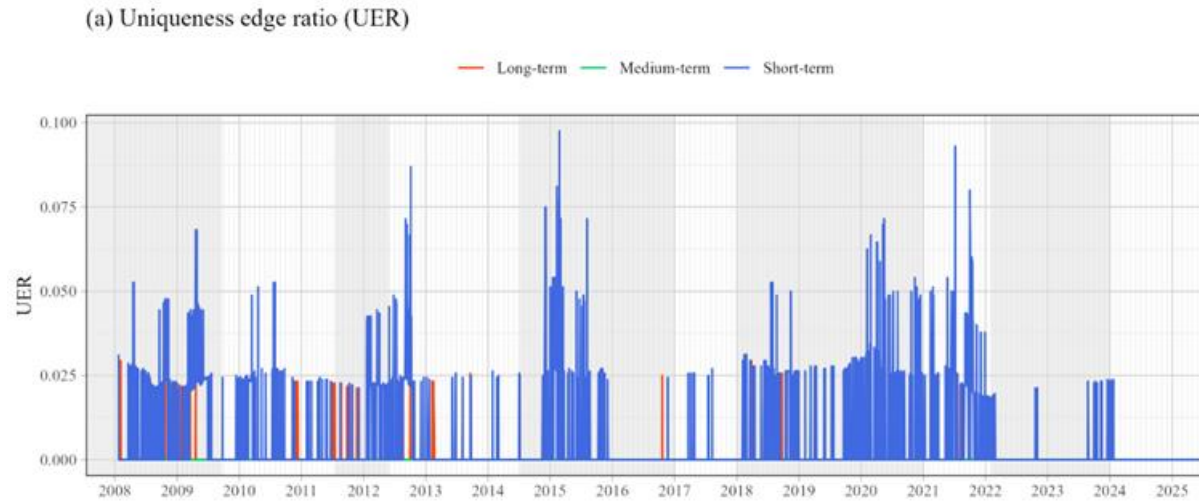
The AND metric looks at something slightly different: it measures the average connectivity of a commodity's neighbours in the network. Essentially, it's telling us about the "reach" each node has and how much potential there is for risk to get amplified as it bounces through connected commodities

- **Trend and Memory:** When we track AND over time, it follows a similar pattern to what we saw with ACS; the short-term AND is consistently the highest and also the most volatile. This backs up what we found earlier: in the short term, the market isn't just more centralized (shown by high ACS), it's also more clustered together. Highly-connected commodities tend to be neighbors with other highly-connected ones creating these dense pockets of connectivity. And here's why that matters: when you have this kind of local density, any shock that enters the system has a much higher chance of getting amplified as it spreads through these tightly-knit groups
- **Comparative Volatility:** Here's an interesting distinction; the AND shows somewhat less long-term volatility (as supported by [Huang 2023](#)) compared to the ACS. What this tells us is important: while the speed at which risk moves through the system (captured by ACS) can change dramatically when crises hit, the underlying structure; the local clustering and immediate connections between highly-connected commodities; stays relatively more stable over longer time periods.

7.4. Uncertainty and Extreme Risk (UER and AEO)

The network's responsiveness to market conditions is further examined by analysing the time-varying **Uncertainty-to-Extreme-Risk (UER)** and the average network connectivity using the **Average Edge Overlap (AEO)**, which is directly related to the average number of nodes in the VaR equation, $O(\text{avg})$.

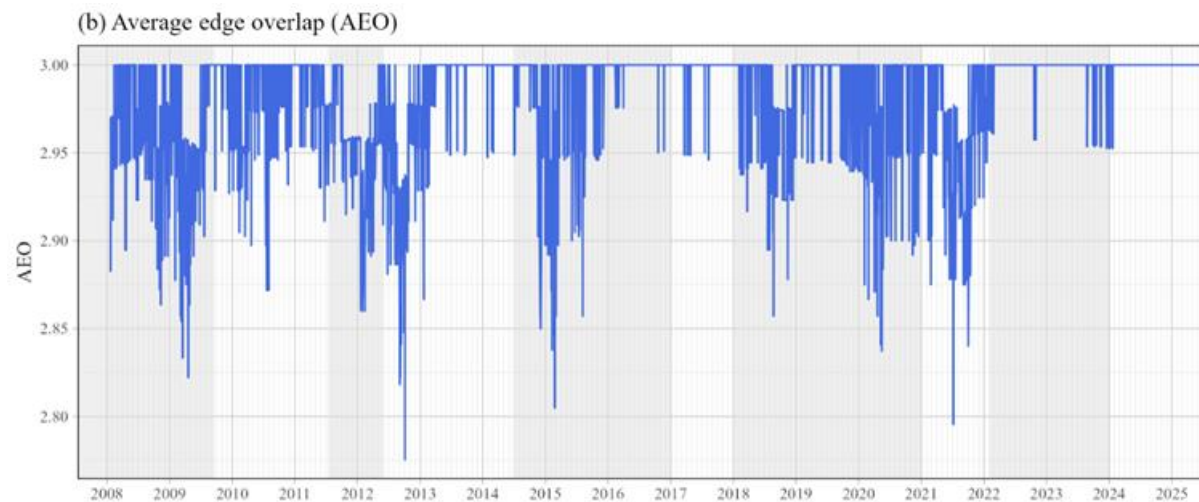
7.4.1 Uncertainty-to-Extreme-Risk (UER)



The UER quantifies the sensitivity of the network structure (in terms of risk linkage edges) to the average number of risk factors, $O(\text{avg})$, used in the individual commodity VaR models.

- **Crisis Sensitivity:** The UER spikes majorly during periods of high uncertainty, most notably during the 2008 GFC and the 2020 COVID-19 crash.
- **Interpretation and Non-Linearity:** These sharp increases display the market's uncertainty forces a rapid, fundamental restructuring of the network. A small increase in the number of required risk factors, $O(\text{avg})$, leads to a disproportionately large, non-linear change in the systemic risk links (edges) of the network (as demonstrated in [barunik 2018](#)). This highlights the unpredictable nature of risk transmission: when panic sets in, a wider array of commodities suddenly become risk factors for others, quickly shifting the network topology and increasing systemic risk.

7.4.2. Average Edge Overlap (AEO / $O(\text{avg})$)



The AEO acts as a proxy for the **Average Number of Nodes $O(\text{avg})$** in the VaR equation.

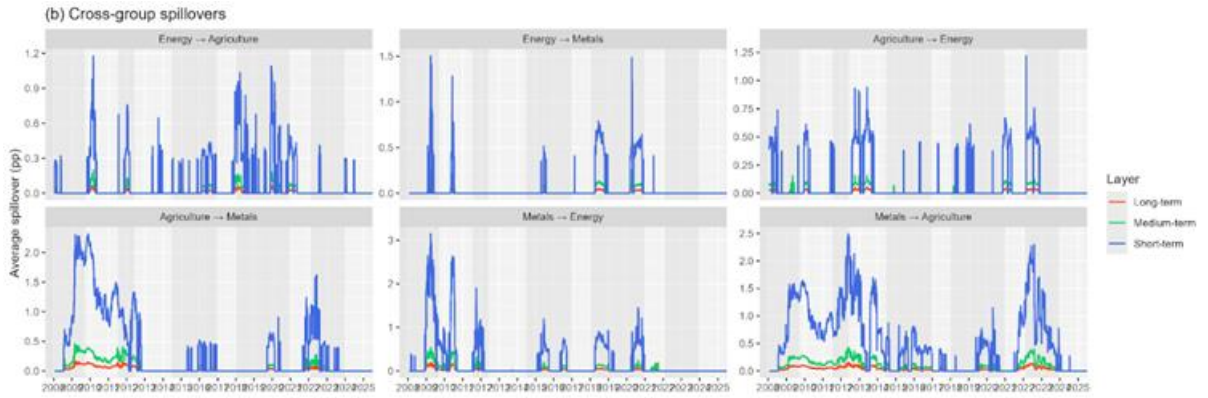
- **AEO Behaviour:** When you track the average over time, you'll see it stays noticeably elevated during major market crises. Both the 2008 Global Financial Crisis and the 2020 pandemic turmoil show significantly higher values compared to calmer periods.
- **Link to Systemic Risk:** This high level of factor overlap tells us something important about how risk works across commodities. When we see elevated AEO, it means that calculating the VaR for individual commodities requires accounting for a bigger, more interconnected set of risk factors that overlap across different markets. What's really happening here is that multiple commodity types are becoming vulnerable to the same triggers at the same time. A single external shock; whether it's a shift in global growth expectations, a change in central bank policy, or another major event; can simultaneously push extreme risk across various commodities. This shared vulnerability is really the heart of why systemic risk spikes in the commodity market when things get stressed. It's not that each commodity is facing its own unique problem; they're all exposed to the same underlying threats.

7.5. Group-Level Connectedness and Risk Transmission

The group-level analysis helps us understand how risk moves within and between the three major commodity clusters; Agriculture, Energy, and Metals.



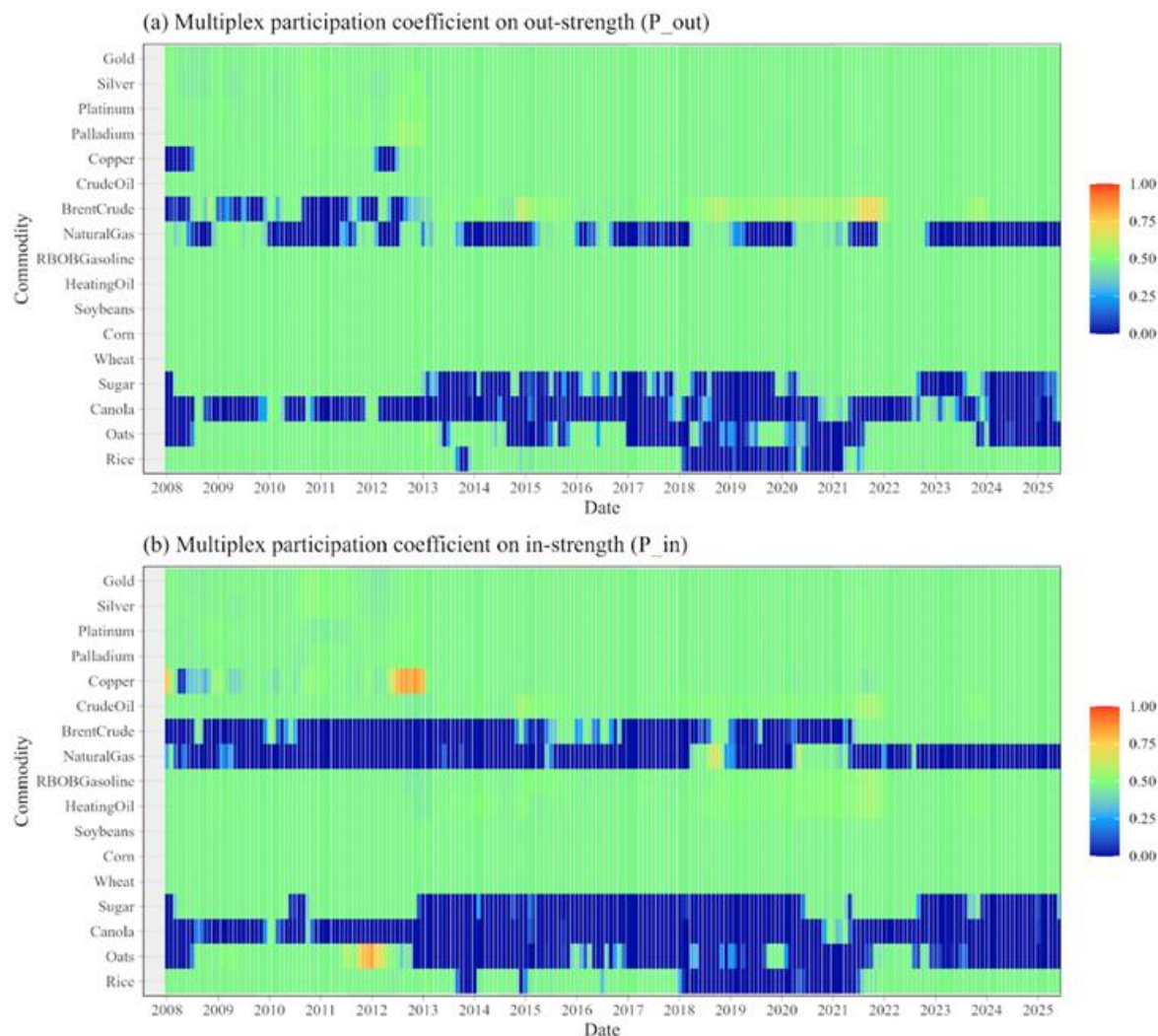
- **Intra-Group Risk (Figure 6a):** When we look at risk connections within each group, Energy stands out as having the highest average intra-group connectedness across all time periods. This confirms what we suspected; the Energy sector is the most systemically risky group, largely because refined and crude oil products are so highly correlated with each other internally.



- Cross-Group Risk Transmission (Figure 6b):** When it comes to how risk moves between different commodity groups, we see a pretty clear pattern; it's mostly one-way traffic, and it's far from balanced. The Energy group acts like a major risk transmitter, consistently pushing risk outward to both the Metals group (EN → ME) and the Agriculture group (EN → AG). This makes sense when you think about it; energy is a fundamental input cost for just about everything. Whether it's fuel for transportation, fertilizer for crops, or power for mining operations, energy costs affect everyone. What's interesting is that the reverse barely happens; risk transmission flowing back from Metals or Agriculture to Energy is consistently low or practically nonexistent. This really reinforces Energy's role as the primary source of shocks in the commodity system.

7.6. Dynamic Systemic Importance (MPC, P(out) and P(in))

The Maximal Path Count (MPC) and the node-level Systemic Importance metrics help us pinpoint which commodities have the most influence and track how stressed the entire system is at any given moment.



- **Maximal Path Count (MPC):** The dynamic MPC (Figure 7) works like a real-time alarm system for systemic risk. It spikes dramatically during major crises; both the Global Financial Crisis and COVID-19 show this clearly. When we see these surges, it's telling us that the network is suddenly filling up with complex, cascading failure pathways. Essentially, the number of potential routes through which contagion can spread multiplies exponentially when the system is under acute stress..
- **Net Risk Roles ($P(out)$ and $P(in)$):** The average $P(out)$ (out-strength) shows us which commodities are net risk exporters; they're sending more risk than they receive. Meanwhile, $P(in)$ (in-strength) identifies the net risk recipients; those on the receiving end.

Rank	Commodity	Avg. P(out)	Rank	Commodity	Avg. P(in)
1	Brent Crude	0.5085	1	Crude Oil	0.4877
2	Palladium	0.4811	2	Heating Oil	0.4873

Dynamic Volatility of Systemic Importance: When we look at how much the monthly P(out) and P(in) values bounce around over time (measured by their standard deviation), we start to see which commodities play consistent roles and which ones are more unpredictable:

Crude Oil barely moves at all in its P(out) standard deviation (0.0001), and that's really telling. It means that no matter what's happening in the market; whether things are calm or chaotic; Crude Oil's role as a risk transmitter stays remarkably stable and consistent. It's always pushing risk outward, regardless of what's happening in the broader market.

Brent Crude tells a completely different story. It shows the highest volatility in its systemic role (0.1841 Std Dev), which really drives home how much Brent's importance can swing depending on what's going on in the world. It's incredibly sensitive to geopolitical tensions and macroeconomic shifts, and its role in the system can change rapidly as market conditions evolve.

8. Discussion

The integrated results from the dynamic multilayer network analysis provide a robust, detailed framework for understanding the mechanisms and evolution of systemic risk within the commodity market over the last two decades.

The simultaneous and extreme peaks across the individual risk measure (LOO), the transmission speed measure (ACS), and the complexity measure (MPC) during crises confirm that systemic instability is a multi-dimensional failure. It is characterized by: (1) High individual vulnerability (driven by the Energy sector), (2) Rapidly increasing contagion speed (high ACS(Short)), and (3) Exponential proliferation of cascade paths (high MPC).

The **rigid risk structure** implied by the high similarity coefficients across short, medium, and long maturity layers is a crucial finding. This suggests that the fundamental economic drivers (e.g., universal energy input costs) impose a uniform structure on the market, limiting the effectiveness of attempts to diversify risk by varying contract duration.

The **Energy group's dominance** as a unidirectional net risk exporter is the most significant structural observation. The Energy sector acts as the **financial accelerator** of the commodity market, efficiently

translating external shocks (financial or geopolitical) into cost-push pressures that propagate into the Metals and Agriculture groups without significant feedback.

The node-level analysis provides a critical distinction between the two core oil benchmarks, defining their unique systemic roles:

- **Brent Crude** is the dominant **dynamic risk driver**, its high volatility in P(out) marking it as the benchmark most susceptible to changing geopolitical regimes.
- **Crude Oil** is the **structural risk node** (highest LOO and P(in), stable P(out)), meaning it is the market's greatest individual risk contributor and its largest recipient of internal Energy shocks, making it the most reliable indicator of persistent systemic vulnerability.

Finally, the relationship between the UER and the AEO/OO(avg) reveals the market's core instability mechanism. The simultaneous elevation of UER and AEO during stress periods indicates that market uncertainty forces a non-linear, fundamental restructuring. Systemic integration occurs because a larger, shared pool of external macro-risk factors (e.g., inflation expectations, central bank actions) simultaneously affects the extreme risk of a wider range of commodities. This collective susceptibility to common factors turns otherwise independent commodity markets into a single, tightly coupled system, explaining why traditional risk models based on stable, historical correlations are likely to fail precisely when they are most needed.

9. Further Research & Conclusion

This study develops and applies a multilayer causal spillover framework to examine systemic risk transmission across global commodity markets over short-, medium-, and long-term horizons. By decomposing causal linkages into frequency-specific layers, we uncover that systemic dependence in commodities is a high-frequency and short-lived phenomenon. The magnitude of causal spillovers decays by more than ninety percent between the one-day and twenty-two-day horizons, confirming that the intensity of systemic risk is concentrated at the shortest timescales. Despite this sharp decay, the network topology remains remarkably stable, as indicated by near-perfect rank correlations across layers. So, crises don't create new connections; they make existing ones stronger within a stable structural core.

The static multilayer analysis shows that Energy and Metals have the most spillovers within a group, while cross-sector contagion is very low. This clustered setup suggests that the commodity system works like a bunch of semi-autonomous risk islands, each of which is dense on the inside but protected on the outside. This confirms traditional diversification strategies across sectors from a portfolio point of view, where being exposed to multiple clusters makes the system less vulnerable.

Dynamic analysis further reveals that the Loss of Overall Connectivity (LOO) acts as a sensitive barometer of market fragility. Peaks in Crude Oil's LOO coincide with known crisis episodes—most notably the COVID-19 shock and the 2020 oil-price war—when its systemic importance surged by nearly one-third. These results indicate that systemic importance is counter-cyclical, increasing

precisely when market resilience diminishes. The accompanying stability of network density reinforces that stress events are characterized by magnitude volatility rather than structural reconfiguration.

The study finds that systemic risk in commodity markets is temporary in time but long-lasting in structure, caused by strong, short-term causal amplification within sectoral clusters. It also needs to use cross-sector diversification to lower long-term exposure. The causal multilayer approach gives us a better way to see how market fragility changes over time and gives us a more timely basis for macro-prudential oversight.

While the multilayer causal spillover model provides new insights into the temporal anatomy of systemic risk, several extensions remain open for exploration. First, future research could incorporate volatility-based or tail-risk measures into the causal network, allowing for the joint modeling of return and volatility spillovers under extreme market states. This would enhance the model's ability to capture nonlinear amplification during crises. Second, expanding the framework to include cross-asset interactions—such as linkages between commodities, equities, and currencies—could reveal whether the sectoral insulation observed here persists at the global financial system level. Third, integrating exogenous macroeconomic variables or geopolitical indicators into the multilayer architecture could help disentangle the drivers of systemic amplification, distinguishing between endogenous market feedbacks and externally induced shocks. Finally, the use of Bayesian or adaptive learning network algorithms may improve causal inference over time, enabling the model to update dynamically as new information emerges.

By pursuing these directions, future studies can deepen the understanding of how systemic risk evolves across assets, horizons, and economic regimes, and thereby extend the practical relevance of the multilayer causal spillover methodology for both investors and policymakers.

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